

HTML Advanced Lab Problems

Text • Formatting • Anchor/Link • Image Elements

Difficulty: Intermediate to Difficult | **Focus:** Practical HTML problem solving

Important: These are lab-oriented problems. Each problem requires students to combine multiple HTML concepts to build a realistic page. The objective is to test understanding and decision-making, not simple tag memorization.

Topics Allowed

- Headings: <h1> to <h6>
- Text: <p>,
, <hr>
- Formatting: , , <i>, , <mark>, <u>, <pre>, <code>, <sub>, <sup>
- Anchor/link: <a> and anchor attributes taught in class
- Images: and image attributes taught in class
- Clickable images using <a> with

Restriction: Do not use CSS, JavaScript, lists, tables, forms, semantic layout elements, Bootstrap, frameworks, or topics outside the current syllabus.

Lab Problem 1: Interactive University Prospectus

Scenario: A university wants a single HTML page that acts like a digital prospectus for prospective students. The page must be understandable even before CSS is introduced.

Your Task

1. Create a meaningful page title and a logical heading hierarchy beginning with <h1>.
2. Create sections for University Overview, Principal/Director Message, Academic Programs, Campus Facilities, Student Life, Admissions, and Contact.
3. Write meaningful multi-paragraph content for every major section. Do not use Lorem Ipsum.
4. Use at least six different text-formatting elements purposefully. At least one sentence must contain nested formatting.
5. Include campus, academic-activity, and student-life images. Each must have meaningful src and alt information.
6. Create links to the university homepage, admission portal, student portal, and an academic resource.
7. Use appropriate anchor attributes taught in class. At least two links should demonstrate different opening behavior.
8. Create a contact block using intentional
 elements.
9. Use <hr> to divide major sections.
10. Make the campus image clickable and connect it to the university homepage.
11. Add one meaningful academic formula using <sup> or <sub>.
12. Include a small technical example using <pre> and <code>.

Difficulty Challenge: Students must decide which heading level, formatting element, link attribute, and image attribute is appropriate instead of adding tags mechanically.

Lab Problem 2: News Portal — Featured Story + Related Stories

Scenario: A college media club needs a breaking-news page containing one detailed featured story and several shorter related stories.

Your Task

1. Create the portal name as the main <h1>.
2. Create a Featured Story with a headline, subheadline, at least five paragraphs, and a featured image.
3. Use for the most important fact, for emphasis, and <mark> for a critical date or statistic.
4. Create at least three Related Story sections. Each must contain a heading, paragraph, and image.
5. Every image must have meaningful alternative text.
6. Each Related Story must contain a Read More link.
7. Make every related-story image clickable and point it to its corresponding article page.
8. Include a source/reference link for the featured story and demonstrate appropriate external-link behavior.
9. Use <hr> between the featured story and related stories.

10. Add an Editor's Note using at least two formatting elements.
11. Include a short code or structured-text example using `<pre><code>`.

Difficulty Challenge: The challenge is to create a clear content hierarchy while making images and links behave like real news-portal navigation.

Lab Problem 3: Travel Guide — One Destination, Multiple Experiences

Scenario: A travel company wants a destination guide for Agra, Jaipur, Delhi, Manali, Goa, or another approved destination.

Your Task

1. Create a destination title using `<h1>` and an introductory paragraph.
2. Create sections for Top Attractions, Historical/Cultural Experience, Food Experience, Best Time to Visit, Travel Tips, and Useful Resources.
3. Include at least four destination images, each representing a different experience.
4. Every image must have meaningful alt text.
5. Create at least four useful external links such as tourism, attraction, transport, or travel-resource pages.
6. Use title attributes on selected links to provide useful context.
7. Make at least two images clickable, with each image leading to a different relevant resource.
8. Use `<strong>` for critical travel advice and `<mark>` for the recommended season.
9. Use `<em>`, `<i>`, or `<u>` only when their use has a clear purpose.
10. Use `<hr>` between major sections and `<br>` for a multi-line contact block.
11. Include a meaningful `<sub>` or `<sup>` example if possible.

Difficulty Challenge: An image should lead to a relevant resource, not simply to a random website.

Lab Problem 4: Product Launch Page — Premium Tech Product

Scenario: A technology startup is launching a fictional laptop, smartphone, smartwatch, or similar product. The first product page must be HTML-only.

Your Task

1. Create the product name as `<h1>` and a persuasive tagline as a lower-level heading.
2. Create Product Introduction, Key Benefits, Technical Information, Pricing, Special Offer, Customer Information, and Learn More sections.
3. Add at least three product images showing different views or use cases.
4. Give every image meaningful alt text.
5. Use formatting elements to distinguish important product information, including `<strong>` and `<mark>`.
6. Display original price, discounted price, and offer information using appropriate formatting.
7. Create Buy Now, Learn More, Specifications, and Support links.
8. Use different anchor attributes appropriately across these links.
9. Make at least two product images clickable, each leading to a different destination.
10. Include a technical specification block using `<pre><code>`.
11. Use `<sup>` and/or `<sub>` where they make sense in technical specifications.
12. Create a support/contact block using `<br>`.
13. Use `<hr>` between product sections.

Difficulty Challenge: The difficulty comes from combining product images, nested anchors, formatted prices, technical text, and multiple link behaviors without CSS.

Lab Problem 5: FINAL LAB — Digital Magazine Home Page

Scenario: You have been hired to create the HTML foundation of a digital magazine for college students covering technology, education, travel, career, and campus life.

Your Task

1. Create a magazine name and main `<h1>`.
2. Create at least five categories: Technology, Education, Travel, Career, and Campus Life.
3. Each category must contain at least one featured article with a heading, multiple paragraphs, an image, and a relevant link.
4. At least two categories must contain two articles, giving a total of at least seven articles.
5. Use a logical hierarchy of `<h1>`, `<h2>`, and `<h3>`.
6. Use at least eight different formatting elements across the page, but never randomly.
7. Every article image must have meaningful alt text.
8. At least four article images must be clickable and each must point to a different article/resource.
9. Create at least six external links and at least two links to hypothetical local HTML pages such as `article.html` or `about.html`.
10. Demonstrate multiple anchor attributes taught in class based on the purpose of each link.
11. Add an Editor's Pick section and highlight its most important phrase using `<mark>`.
12. Add a Quote/Editor's Comment section using suitable formatting elements.
13. Include one technology article containing a `<pre><code>` block.
14. Include one article containing at least two meaningful uses of `<sup>` or `<sub>`.
15. Use `<hr>` between major categories and `<br>` only for intentional line breaks.
16. Do not use CSS, JavaScript, lists, tables, forms, semantic layout tags, or untaught topics.

Difficulty Challenge: This is the hardest lab. Students must design the complete information hierarchy themselves and demonstrate when each element should be used.

Lab Completion Checklist

- Standard HTML document structure is present.
- Heading hierarchy is logical.
- Content is meaningful and scenario-specific.
- Formatting elements are used purposefully.
- Images have working sources and meaningful alternative text.
- Links point to relevant destinations and use appropriate attributes.
- Clickable images are correctly created using anchor + image.
- hr and br are used intentionally.
- pre and code are used correctly where required.
- No CSS, JavaScript, lists, tables, forms, or untaught topics are used.